



Efficient 3D face recognition handling facial expression and hair occlusion[☆]

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ABSTRACT

This paper presents an efficient 3D face recognition method to handle facial expression and hair occlusion. The proposed method uses facial curves to form a rejection classifier and produce a facial deformation mapping and then adaptively selects regions for matching. When a new 3D face with an arbitrary pose and expression is queried, the pose is normalized based on the automatically detected nose tip and the principal component analysis (PCA) follows. Then, the facial curve in the nose region is extracted and used to form the rejection classifier which quickly eliminates dissimilar faces in the gallery for efficient recognition. Next, six facial regions which cover the face are segmented and curves in these regions are used to map facial deformation. Regions used for matching are automatically selected based on the deformation mapping. In the end, results of all the matching engines are fused by weighted sum rule. The approach is applied on the FRGC v2.0 dataset and a verification rate of 96.0% for ROC III is achieved as a false acceptance rate (FAR) of 0.1%. In the identification scenario, a rank-one accuracy of 97.8% is achieved.

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1. Introduction

Face recognition is usually required in a wide range of applications such as surveillance, access control and machine–human interaction because of its non-intrusive nature. 2D face recognition has been extensively researched in the last two decades, but it is sensitive to illumination and pose. Therefore, researchers are now investigating other data acquisition modalities of the face to overcome these limitations. One of the most promising modalities is the 3D shape of the face. The 3D face data is considered to be inherently robust to pose and illumination which adversely affect 2D face data [1].

The 3D shape of the human face will be sharply deformed under facial expressions in a complex way which result in deterioration in the accuracy of recognition. It is noted that the expression variation is one of the most difficult factors in 3D face recognition.

There also exist some attempts to overcome the expression variation in the 3D face recognition field. Bronstein et al. [2] demonstrate that facial expressions can be modeled as isometries of the facial surface. They assume the 3D facial surface to be isometric and remove the effects of expressions by finding an isometry-invariant representation of the face. Lu et al. [3] learn expression deformations from a control group of neutral and non-neutral scans. Then, the learned deformations are used to synthesize an expression model for the neutral models in the gallery. The comparisons between the probe and the deformed galleries are performed to finish the recognition. In the

work by Kakadiaris et al. [4], the 3D face is parameterized using the annotated face model (AFM). In the work by Wang et al. [5], fields are applied to deform the non-neutral 3D scans to neutral scans and meanwhile constraints are applied to avoid overfitting. Al-Osaimi et al. [6] propose a 3D face recognition approach which differentiates between expression deformations and interpersonal disparities using deformable models.

Different parts of the face deforms differently when expression changes: some regions deform largely and others little. Using rigid regions which are relatively less sensitive to expressions is another way to overcome the expression variation in 3D face recognition. Chang et al. [7] use multiple overlapping nose regions and obtain increased performance relative to using the whole-frontal-face region. Mian et al. [8] describe a multimodal 2D + 3D face recognition system. In their system, three kinds of methods, spherical face representation (SFR), scale-invariant feature transform (SIFT) descriptor and a region-based matching method, are combined. The system improves the efficiency of recognition by using SFR and SIFT to reject a subset of gallery subjects, and receives promising results, but some information sensitive to expressions is used to produce SFR which will deteriorate recognition. Both methods above, however, don't make full use of the discriminative information existing in the entire face. Queirolo et al. [9] use a simulated annealing-based approach (SA) for range image registration, and then compute the similarity between two faces by surface interpenetration measure (SIM). Four regions are used for matching and the matching classifier is based on a hierarchical model. Significant recognition accuracy is reported for that approach.

In the work by Wang et al. [10], the rigid parts of the facial surface are dynamically extracted by selecting a part of the nearest point

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pairs to calculate dissimilarity measure during registration of facial surfaces. Faltemier et al. [11] select 28 different regions around the face, and then perform score-based fusion on the individual region match scores. A hierarchical component-based face registration method is used in [12]. In the work by Lin et al. [13], all of the matching points are weighed by the distance from itself to the bilateral symmetrical plane of the face. The methods above use iterative closest point (ICP) [14] to perform image matching, and achieve good performance, but the time consumption is large because all of the face regions should be matched in the recognition process. Ocegueda et al. [15] proposed a novel framework to find the most discriminative areas of the face for recognition by a Markov random field model. For each 3D face, a compact signature consisting of 360 coefficients was extracted and used for recognition. Contrary to our approach, this method requires mapping to obtain a geometry image for each face.

Considering that face is usually deformed in different ways under different expressions, some researchers combine facial expression categorization with face recognition. In the work by Cook et al. [16], images are manually allocated to three groups based on expression, and then a part-face recognition system using subspace projection method is constructed. However, the above method assumes that the expression of the face is known. In [17], an automated variant of

this approach is detailed, and a “happy face” recognition system is used to aid the face recognition.

Beumier and Acheroy [18] develop an integrated 3D face acquisition and comparison system based on multiple-profile comparisons. The facial curves are extracted and used to compare two faces. There are also many other works employing the facial curves for 3D face recognition [19]. It is reported that the central profile shows the major distinctiveness [20]. Generally, the facial curves are less discriminative than the regions, but they only need low storage cost and have high speed.

Another factor that usually deforms the face is hair occlusion, and hair often occludes the forehead region. In this paper, we present a robust 3D face recognition method handling facial expression and hair occlusion. A complete framework for face recognition is presented and it has five main stages (see Fig. 1): (1) preprocessing, (2) rejection, (3) adaptive region selection, (4) region-based matching, and (5) evaluation.

In the preprocessing stage, the input face is firstly smoothed and the nose tip is automatically detected. Then, the face is cropped and normalized. The facial curve in the nose region which is also called “the partial central profile” is matched to reject dissimilar faces in the gallery. Curves in the forehead, cheeks and mouth regions are extracted and matched to produce a facial deformation mapping for

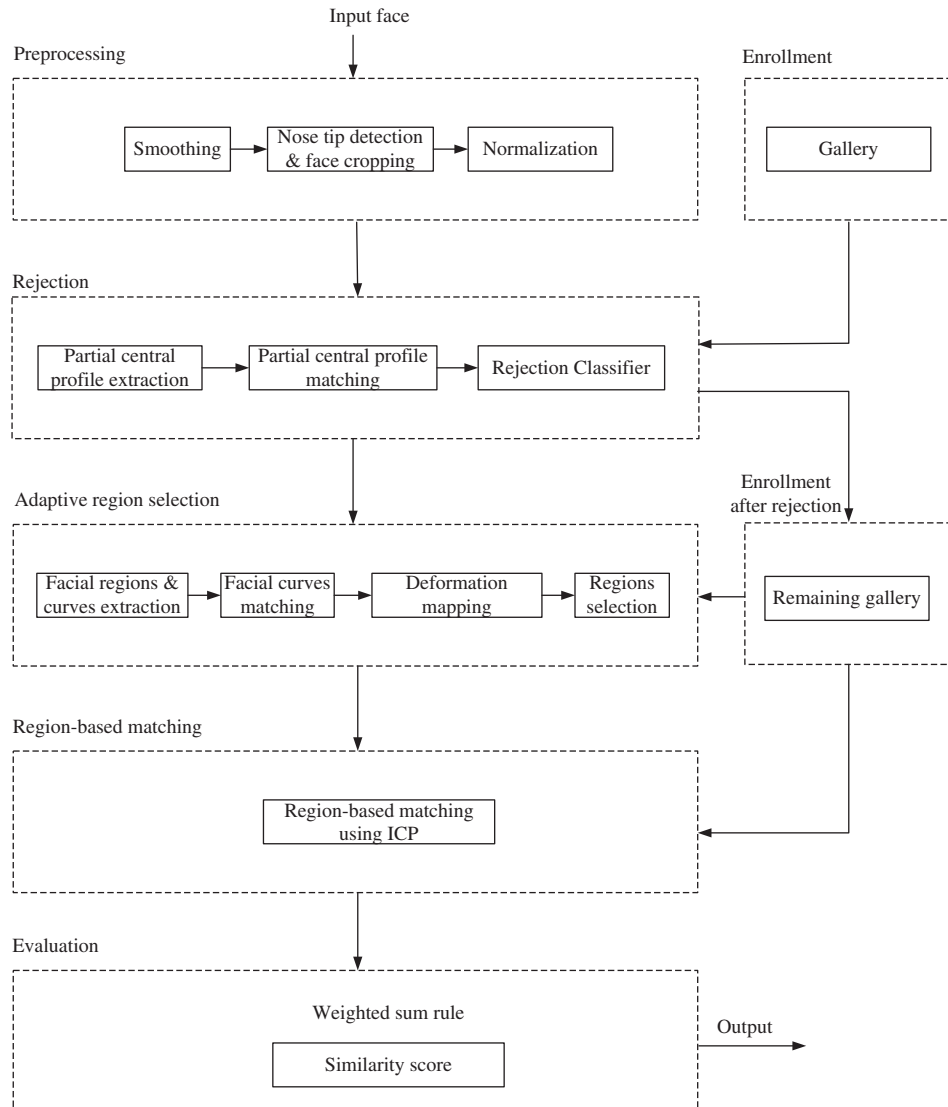


Fig. 1. Block diagram of our algorithm.

expression and hair occlusion. Regions used for matching are adaptively selected based on the deformation mapping. The selected regions are matched to the remaining gallery using a modified ICP algorithm. Lastly, similarity score is computed based on the weighted sum of elementary scores given by each region and the face with the highest similarity score is considered as the recognition result. Fig. 1 shows the block diagram of our method.

3D face recognition is computationally expensive, and a brute force matching approach does not scale well to large galleries. AdaBoost is used to select a small number of critical features in 2D face detection [21], and there are also researchers using a cascade method for fast human detection [22]. In this paper, a rejection algorithm is used to reduce the gallery faces before accurate matching. A rejection classifier can reduce the number of galleries before accurate matching, but what feature is used for rejection is a key problem, which is the first contribution of our paper. We use the partial central profile to produce a rejection classifier. Partial central profile is a vertical facial curve in the nose region. It is simple and robust to facial expression and hair occlusion, so it can reject a large number of ineligible candidate faces in a short time.

The second contribution of our paper is an adaptive region selection scheme. After rejection, our scheme can automatically select as much information as possible for accurate matching, and the information is robust to expression and hair occlusion. Facial curves in the regions of forehead, cheeks and mouth are extracted and compared to produce the facial deformation mapping, then regions used for matching are selected according to the deformation mapping. Through looking at the distortions on a region by region basis, our scheme can scale to any number of facial expressions without requiring the use of expression recognition systems as in [17]. Comparing with the work by Wang et al. [10] and Faltemier et al. [11], only regions that are not influenced by expression or hair occlusion are matched and used for recognition in this paper, which can improve the efficiency. This also has the benefit over systems such as those in [7,8], where only the nose and forehead regions are used for recognition.

This paper is an extension of our work presented in [23] and is organized as follows. Section 2 describes the preprocessing stage. In Section 3, the rejection classifier based on the partial central profile is presented. Section 4 gives details of our adaptive region selection scheme. Section 5 lists and compares the recognition results of our algorithm with others. Conclusions are discussed in Section 6.

2. Face preprocessing

2.1. Face smoothing

We perform experiments on the Face Recognition Grand Challenge (FRGC) v2.0 dataset [24]. Our algorithm operates automatically by

using only the 3D shape from the frontal view of the face. There are outlier and missing points in many images in the FRGC v2.0 dataset, which will deteriorate the recognition performance, so the images are firstly smoothed. Spikes caused by outlier points are removed by Gaussian filter. Then, small holes in the face are filled. The X, Y and Z coordinates of the missing points are interpolated from its valid neighbors.

2.2. Nose tip detection and face cropping

Since faces in the FRGC v2.0 dataset are mostly acquired from the shoulder level up (see Fig. 2(a)), the faces should be localized. It is well known that the location of the nose is around the centroid of the face, so the nose tip is firstly detected to crop out the required facial area.

The nose tip is detected as follows. At first, local curvature information at each point, named shape index, is calculated. Shape index is proposed by Dorai and Jain [25], and it represents the local shape information of the points. Shape index at point p is calculated using the maximum ($\kappa_{\max}(p)$) and minimum ($\kappa_{\min}(p)$) local curvature as

$$\text{shape index}(p) = \frac{1}{2} - \frac{1}{\pi} \tan^{-1} \frac{\kappa_{\max}(p) + \kappa_{\min}(p)}{\kappa_{\max}(p) - \kappa_{\min}(p)}. \quad (1)$$

Shape index is independent of the coordinate system, and can scale the shape of the surface. A value of zero represents a spherical cup, while a value of one represents a spherical cap. Considering that the nose region is a convex region, so points whose shape index is in the range of 0.85–1 are grouped to form possible nose tip regions. Since there may be small regions caused by hair or other factors, small regions (<70 points) are removed. After this, the possible nose tip regions are labeled as $S1_{nt}$. Note that these regions include areas around the ears, eyes, lips and chin as well. After that, following [27], two nose tip templates are used. First, a smaller nose tip template is used to match the face to determine which regions in $S1_{nt}$ locally resemble a nose tip. Then, for the regions where the smaller nose tip template fits well, we fit a larger template with the nose tip being marked to select the actual nose tip p_{nt} .

After the nose tip p_{nt} is determined, a sphere of radius r centered at p_{nt} is then used to crop the 3D face. A constant value of $r=90$ mm is selected in this paper. This process crops an ellipsoid region, labeled R1, as shown in Fig. 2(b).

2.3. Face normalization

Principal component analysis (PCA) [26] is used to normalize the face pose. As shown in Fig. 2(b), it can be found that the cropped facial region is larger from the vertical orientation of the face than that from the horizontal orientation, and the thickness between the

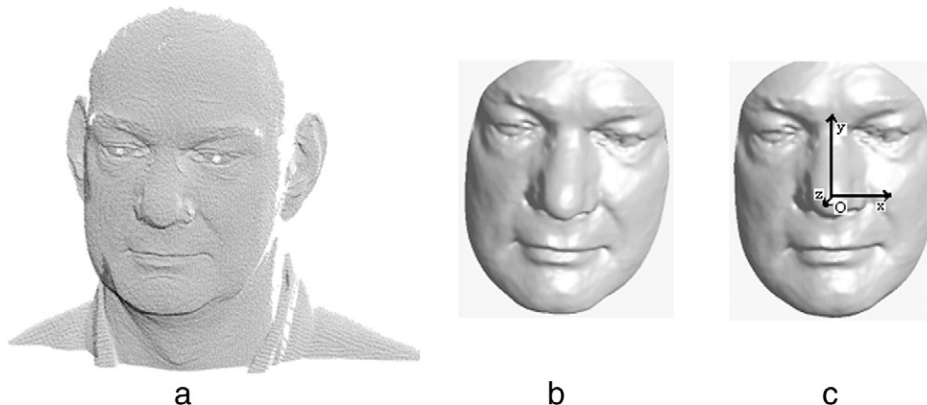


Fig. 2. Face cropping and normalization. (a) A pointcloud of an original face, (b) the cropped face after smoothing, and (c) the face after normalization.

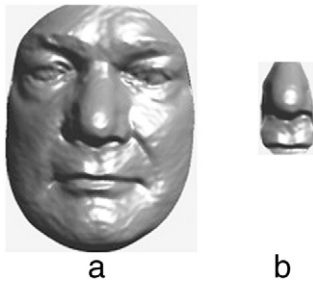


Fig. 3. Facial regions used for normalization. (a) The region R1 of a sample 3D face, and (b) the region R2 of a sample 3D face.

pre- and post-face is the smallest. The shape of the facial region is approximately ellipsoid. Hence, by PCA on the cropped face, three eigenvectors can be obtained with their corresponding eigenvalues. Taking the eigenvector corresponding to the largest eigenvalue as the Y axis, and the eigenvector corresponding to the smallest eigenvalue as the Z axis, we define a new right-hand coordinate system. This coordinate system is called pose coordinate system (PCS) [20], and all faces have the same front view in the PCS as shown in Fig. 2(c). The facial surface is then transformed to the PCS and the origin of PCS is transformed to the nose tip p_{nt} . Generally, almost all the points of the face in PCS have a negative z value. The cropped face region R1 (see Fig. 3(a)) is prone to be affected by hair and the pose error along the Z axis will occur when faces have a small aspect ratio (major axis to minor axis ratio). Therefore, after the pose normalization process above, a smaller region is cropped from the face using a radius of 50 mm centered at the nose tip and a z value threshold equal to -25 mm. This results in a new region with a higher aspect ratio (see Fig. 3(b)), labeled R2, and the new region R2 is used to normalize the face for the second time.

Fig. 4 shows the sample faces before and after normalization. It is important to note that we obtain a more accurate pose with the smaller region. In the first row in Fig. 4, compared with the pose corrected by only the region R1, the Y axis is more accurate when both R1 and R2 are used, and the Z axis is tuned when region R2 is used in the second row. The reason is that the smaller region is generally influenced less by hair, and we have a higher aspect ratio which will be beneficial to obtain a more accurate Z axis. After this, we register the face to a normal face using ICP algorithm to obtain a

more accurate normalization. The normal face is pose corrected during offline processing, and the face will have the same pose as the normal face after registration.

The majority of the images causing this algorithm to fail are due to hair artifacts in the large region and exaggerated expressions. Visual analysis of the nose tip detection algorithm applied to the 400 scans randomly selected from the FRGC v2.0 dataset determines that only 1 of the nose tip is more than 2 mm away from the manually marked nose tips. All the 400 scans were normalized using our method and the corresponding PCS were produced. There were 398 PCS which were less than 1° away from the normal PCS. And the normalized PCS was produced by comparing the input face to a normal face with the PCS being manually marked.

3. Rejection classifier based on the partial central profile

3.1. Rejection classifier

A rejection classifier is defined as the one which quickly eliminates a large percentage of the candidate classes with high probability. A rejector is an algorithm, ψ , that given an input, $x \in \mathcal{R}^d$, returns a set of class labels, $\psi(x)$, such that $x \in W_i \Rightarrow i \in \psi(x)$, where x is a measurement vector, $\mathcal{R}^d$ is a classifier space of d measurements, and W_i is the i th class such that $W_i \subset \mathcal{R}^d$ [8,28].

The effectiveness of a rejection classifier, ψ , can be defined as

$$Eff(\psi) = \frac{E_{x \in S}(|\psi(x)|)}{N}, \quad (2)$$

where $E_{x \in S}(|\psi(x)|)$ is the expected cardinality of the rejector output, N is the number of input vectors, and it is the size of the gallery in this paper. The smaller the value of $Eff(\psi)$, the better is the rejection classifier. In general, while it significantly reduces the number of possible classes, a rejection classifier does not provide a final solution to the pattern recognition problem. Consequently, the recognizer can dedicate its computational resources to a much smaller number of candidates, so the efficiency of recognition is improved.

3.2. Rejection based on the partial central profile

In order to obtain good recognition performance, there are mainly two requirements for the features used for the rejection classifier:

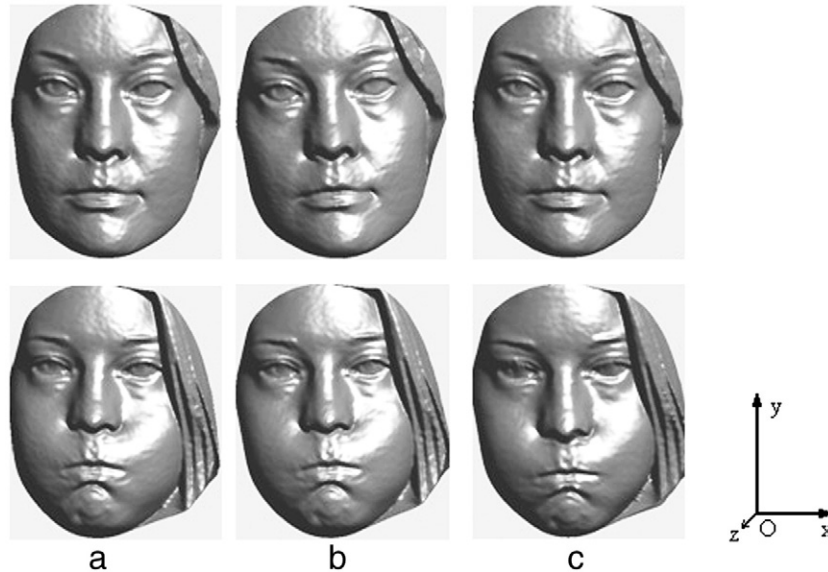


Fig. 4. Sample 3D faces before and after pose correction. (a) 3D faces before pose correction, (b) corresponding 3D faces after pose correction using only region R1, and (c) corresponding 3D faces after pose correction using both regions R1 and R2.

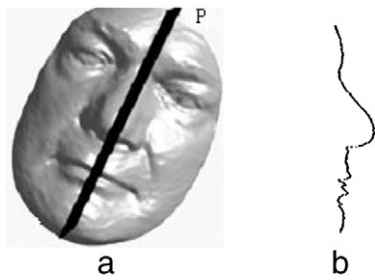


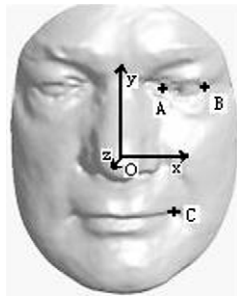
Fig. 5. The definition of the facial curve. (a) The intersection of the plane and facial surface, and (b) intersection line.

(1) the features must be robust, which will not reject the eligible candidate faces when they have different expressions or hair occlusion; and (2) the features must be simple, which can reject a large number of dissimilar candidate faces in a short time. It is well known that the nose is the most robust region in the face and facial curves can represent the distinctiveness of the face and are low cost. So the vertical facial curve in the nose region (partial central profile) is selected to form the rejection classifier in this paper.

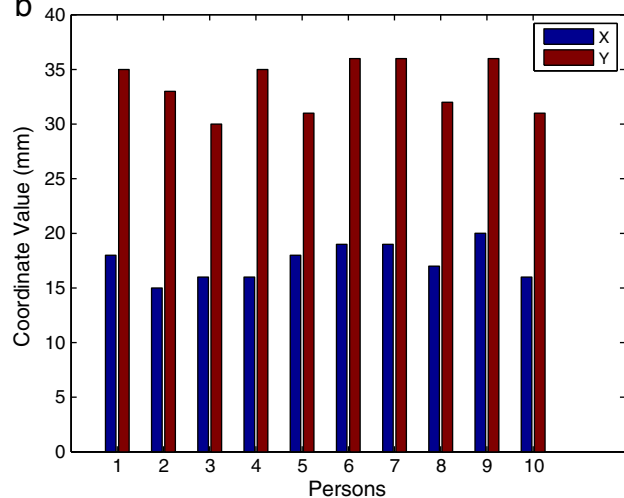
A facial curve is defined as the intersection of a plane with the facial surface. Fig. 5 illustrates the definition of the facial curve. In

Fig. 5(a), a plane P and the facial surface are intersected, and the intersection line is defined as a facial curve which is shown in Fig. 5(b). Hence, different planes will produce different facial curves. The intersection plane can be determined by feature points such as corners of mouth, upper end of eyebrows and so on [29]. However, it is difficult to accurately detect the feature points in a 3D face. In order to investigate the geometric distribution of feature points in a 3D face, we analyze the coordinate information of feature points with different persons. 10 persons are randomly selected from the training partition of the FRGC v2.0 dataset and the feature points (e.g. eye corners and mouth corners) are manually marked after the face is normalized. One face for each person is used. Then, the X and Y coordinate information of the feature points of different persons in PCS is compared (see Fig. 6). Since almost all the faces are symmetrical [19], only the feature points in the right side of the face are analyzed. The feature points are shown in Fig. 6a, the coordinate information of the feature points of different persons are presented in Fig. 6(b), (c) and (d) respectively. From Fig. 6, it can be seen that the coordinate values of the same feature points among different persons in PCS are similar. The geometric distributions of the facial features are almost the same although they are extracted from different persons. Hence we extract the intersection plane based on the coordinate information. The training partition of the FRGC v2.0 dataset is used during the offline process to obtain the coordinate range for

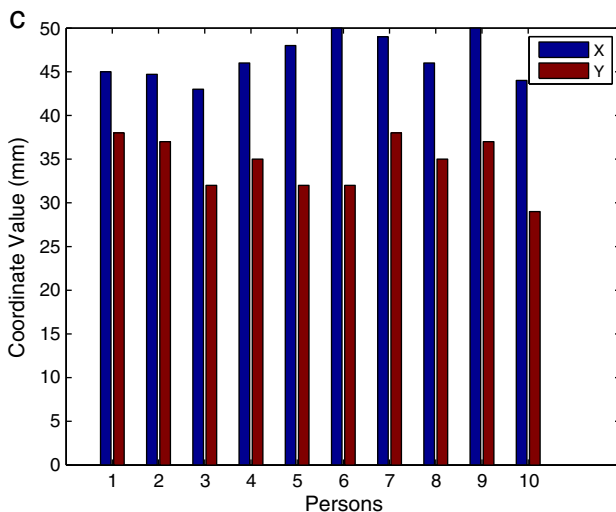
a



b



c



d

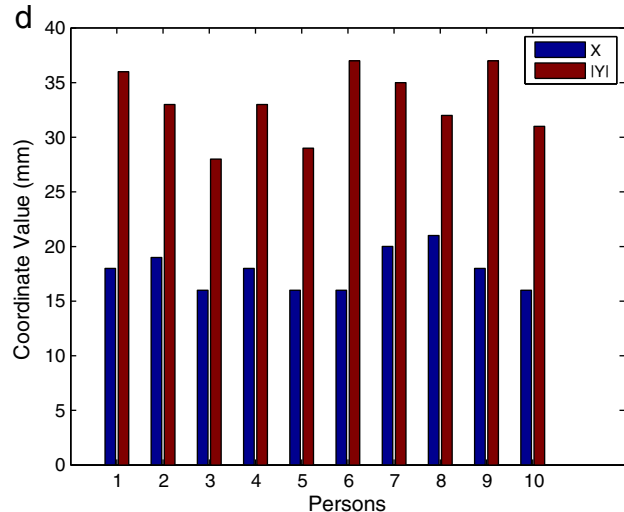


Fig. 6. The coordinate information of feature points with different persons. (a) The feature points A, B, and C, (b) the coordinate information of feature point A, (c) the coordinate information of feature point B, and (d) the coordinate information of feature point C.

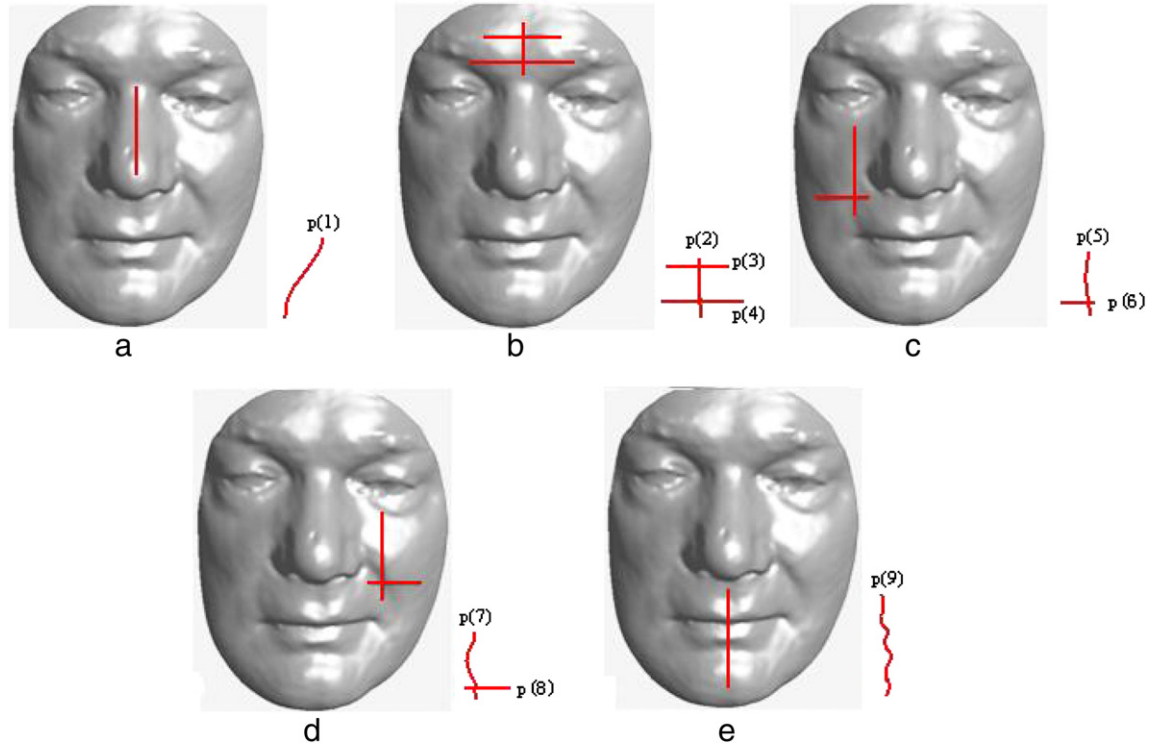


Fig. 7. Facial curves. (a) facial curve $p_{(1)}$, (b) FACIAL curves $p_{(2)}, p_{(3)}, p_{(4)}$, (c) facial curves $p_{(5)}, p_{(6)}$, (d) facial curves $p_{(7)}, p_{(8)}$, and (e) facial curve $p_{(9)}$.

the plane. The partial central profile $p_{(1)}$ is extracted based on the co-ordination information and is shown in Fig. 7(a), and the profile is also displayed in a non-frontal view on the lower-right corner. The coordinate information is shown in Table 1. X and Y determine the coordinate range of facial curves in the XOY plane of PCS. $p_{(1)}$ is mainly on the region of the nose tip to the eyes. In this paper, the other facial curves used for matching are also extracted in the same way.

3.3. Analysis and result of rejection

Intuitively, the partial central profile representation appears to be less descriptive compared to the entire profile. However, it is less sensitive to facial expression and hair occlusion. Moreover, in terms of computational complexity, the entire profile representation is more complex than the partial profile, because there are more points. Therefore, the partial central profile should be a better choice for an efficient rejector. Fig. 8 graphically illustrates the comparisons between the partial central profiles and the entire central profiles. It can be found that the partial central profiles are more stable.

Once the partial central profile is extracted, matching is performed using ICP algorithm. ICP establishes iteratively correspondences between the closest points of two sets of 3D points and minimizes the distance error between them by applying a rigid transformation to one of the sets. The similarity $s_{p_{(1)}}^i$ is measured by the average distance $d_{p_{(1)}}^i$ between two curves

$$d_{p_{(1)}}^i = \frac{1}{n_p} \sum_{i=1}^{n_p} \|g'_i - p_i\|, \quad (3)$$

where n_p is the size of the points in the matching partial central profile, p_i is the i th point in the matching partial central profile, and g'_i is the corresponding point of p_i in another matching profile after rigid transformation.

In face recognition, face images of known persons are initially enrolled into the system. This set of persons is referred to as the “gallery”. Later images of these or other persons are used as “probe” to match against images in the gallery [1]. The partial central profile of an image in the probe is matched to all the galleries, and the matching process results in a vector of similarity scores $s_{p_{(1)}}$ of size N , where N is the size of the gallery. The scores are normalized to a scale of 0 to 1 (0 being the best similarity) using

$$s_{p_{(1)}}^i = \frac{s_{p_{(1)}}^i - \min(s_{p_{(1)}})}{\max(s_{p_{(1)}}) - \min(s_{p_{(1)}})}, \quad (4)$$

where, $s_{p_{(1)}}^i$ is the i th element of the vector $s_{p_{(1)}}$.

Gallery faces whose similarity score is above a threshold δ_{rej} are rejected. The remaining gallery faces are then accurately matched for the final recognition. Selecting a threshold δ_{rej} is a tradeoff between efficiency and accuracy. In this paper, we obtain the threshold δ_{rej} by the training partition of the FRGC v2.0 dataset. One neutral face for each person in the training set is selected as the gallery, and the remaining faces are selected as the probe. Then, each face in the probe is matched to the gallery faces using partial central profile. Next, several values are selected for δ_{rej} , and for each δ_{rej} , the $Eff(\psi)$ and verification rate can be obtained. In the end, a threshold $\delta_{rej} =$

Table 1
Facial curves information (mm).

	$P_{(1)}$	$P_{(2)}$	$P_{(3)}$	$P_{(4)}$	$P_{(5)}$	$P_{(6)}$	$P_{(7)}$	$P_{(8)}$	$P_{(9)}$
X	0	0	$[-20, 20]$	$[-25, 25]$	-30	$[-55, -25]$	30	$[25, 55]$	0
Y	$[0, 60]$	$[50, 75]$	70	50	$[-20, 25]$	-12	$[25, 55]$	-12	$[-70, -10]$

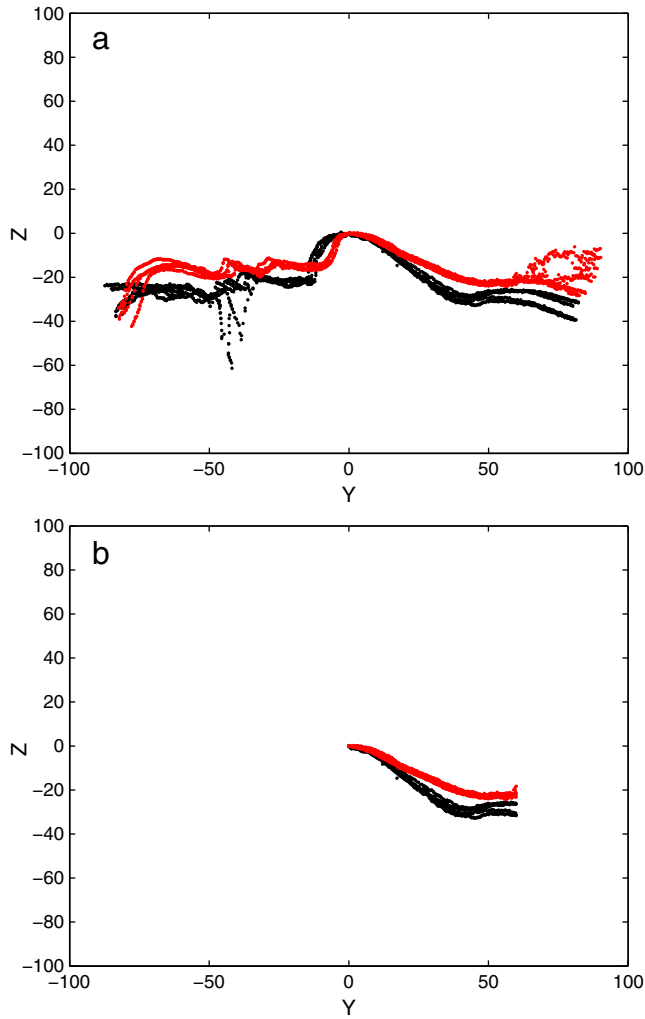


Fig. 8. The comparisons between the partial central profiles and entire central profiles. (a) Entire profiles of two individuals under different expressions and hair occlusion, and (b) partial central profiles of two individuals under different expressions and hair occlusion. The different colors represent different individuals.

0.25 is chosen so that $Eff(\psi) = 0.31$, and the verification rate of the rejection classifier is 98.7%.

4. Adaptive region selection scheme

4.1. Part-face methodology

Face recognition techniques typically employ a monolithic representation of the face during recognition, however, approaches which decompose the face into sub-regions have shown considerable promise [7,8]. Based on the analysis of facial action under expressions [30], we divide the face into six regions which are respectively the nose (Probe A), forehead (Probe B), left cheek (Probe C), right cheek (Probe D), left mouth (Probe E) and right mouth (Probe F) (see Fig. 9). The six regions are selected because each of them usually deforms differently in different cases. Probe A is an isosceles trapezoid and the coordinate information is shown in Table 2. X and Y determine the coordinate range of the nose region in the XOY plane. The cropping parameters of the other five regions can be seen in Table 3. OffsetX and OffsetY determine the new sphere center in relation to the origin. SphereRadius determines the new sphere radius. In order to reduce the calculation time of matching, the gallery faces are

segmented during offline processing. Example of the matching region of the gallery can be seen in Fig. 9(g). The regions used for matching are adaptively selected based on the deformation mapping which will be introduced below in detail.

4.2. Deformation mapping

Facial curves can be thought of as nonuniform sampling of the face, so the deformation of the facial curve can express the deformation of the region which contains the facial curve. In this paper, facial curves are extracted and matched to evaluate the deformation of the face. In each region, facial curves are extracted in or near the parts which are most sensitive to expression or hair occlusion. Finally four vertical and four horizontal curves which are located in the regions of the forehead, cheeks and mouth are extracted (shown in Fig. 7(b), (c), (d), (e)). A vertical curve is extracted as an intersection of a face surface mesh with a plane parallel to the YOZ plane. A horizontal curve is extracted as an intersection of a face surface mesh with a plane parallel to the XOZ plane. The coordinate information of facial curves is shown in Table 1.

The eight facial curves of the probe are matched to the remaining gallery faces using ICP, and for each gallery face eight similarity scores $s_{p(i)}$, ($i = 2, 3, \dots, 9$) can be obtained. The facial deformation can be evaluated using the matching results of the facial curves. All the similarity scores have a negative polarity in this paper, that is, a larger value of $s_{p(i)}$ means a lower similarity. We assume that the higher the value of $s_{p(i)}$, the larger is the deformation presented in the region in which the i th curve is located. There is usually more than one curve for each region, and the maximum $s_{p(i)}$ in every region is chosen to evaluate the corresponding region deformation. Deformation of the nose region can be evaluated by the partial central profile, hence the deformation of all the six regions can be obtained by

$$d_{\text{Probe A}} = s_{p(1)}, \quad (5)$$

$$d_{\text{Probe B}} = \max(s_{p(2)}, s_{p(3)}, s_{p(4)}), \quad (6)$$

$$d_{\text{Probe C}} = \max(s_{p(5)}, s_{p(6)}), \quad (7)$$

$$d_{\text{Probe D}} = \max(s_{p(7)}, s_{p(8)}), \quad (8)$$

$$d_{\text{Probe E}} = \max(s_{p(6)}, s_{p(9)}), \quad (9)$$

$$d_{\text{Probe F}} = \max(s_{p(8)}, s_{p(9)}). \quad (10)$$

Through Eqs. (5) to (10), we obtain the deformation mapping of the face. The larger the $d_{\text{Probe M}}$ is, the more deformation exists in the region Probe M, where $M \in (A, B, C, D, E, F)$.

4.3. Adaptive region selection scheme

Given that the deformation of the different regions of the face caused by expression and hair occlusion can be obtained through the facial curves, we propose an adaptive region selection scheme for face recognition.

Since the nose region is almost invariant to all expressions and generally hair occlusion will not deform it, Probe A is always selected during the accurate matching process. Other regions are selected only when the corresponding deformation $d_{\text{Probe } *}$ is less than a threshold δ_{deform} .

In our method, if faces used for matching belong to the same identity, the deformation $d_{\text{Probe } *}$ is usually caused by expression, hair occlusion and so on (referred to as intra-personal deformation). Hence

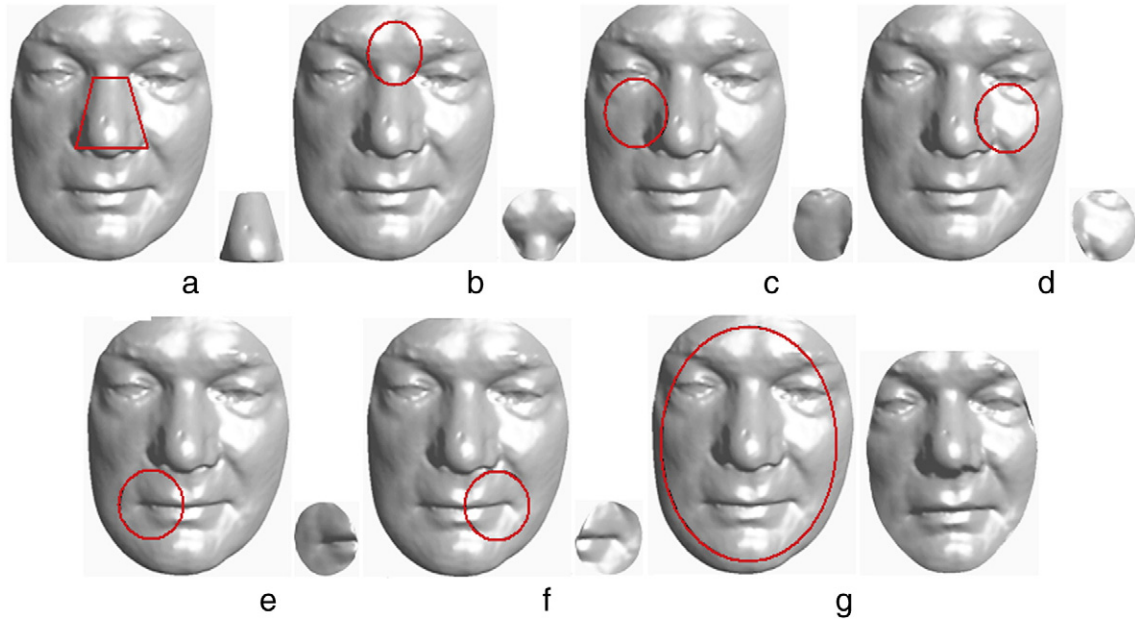


Fig. 9. Matching regions for probe and gallery surface. (a) Probe A, (b) Probe B, (c) Probe C, (d) Probe D, (e) Probe E, (f) Probe F, and (g) a gallery surface.

regions which are not deformed will be selected. However, if faces used for matching belong to different identities, the deformation d_{Probe_s} is constituted of intra-personal deformation and the difference between persons (referred to as inter-personal deformation), and in this situation there are usually two cases for one region to be selected: (1) similar faces with small or no intra-personal deformation; and (2) dissimilar faces with certain intra-personal deformation. The latter will deteriorate the recognition performance when the inter-personal deformation can be completely compensated by intra-personal deformation, but it is a rare case.

To illustrate the results of our adaptive region selection scheme, we register faces with a neutral or non-neutral expression to their respective 3D faces with a neutral expression (see Fig. 10(a), (b), (c), (d)), and faces with hair or mustache occlusion to their 3D faces without occlusion (see Fig. 10(e)). Also, we register two faces which belong to different identities (see Fig. 10(f)). The regions which are selected by our adaptive region selection scheme are labeled with red curves in the bottom row of Fig. 10. We can see that when the matching faces belong to the same identity, our method can select the regions which are not deformed by expressions without knowing the expression of the face. Regions occluded by hair or mustache can be automatically discarded. When the matching faces belong to different identities, the nose region and other regions which are similar in shape are selected for recognition. By identifying faces in which deformation is presented, the recognition process can place more emphasis on those regions which are least affected.

In this paper, for each region the same threshold δ_{deform} is selected. It is true that a vector threshold with values specific for each region will result in better performance. But in fact, the improvement is small. Regions used for matching are selected based on the deformation mapping on the corresponding region no matter what the region is. Except for the nose region, each region whose deformation is larger than δ_{deform} is discarded.

Table 2
Probe A information (mm).

Probe A	Shorter base	Longer base
X	[−20, 20]	[−40, 40]
Y	35	−5

4.4. Face matching and results fusion

Once regions are selected, the ICP algorithm is run on each probe of the remaining galleries. Given a pair of surface to be matched, the initial registration is performed by face normalization. ICP is time consuming because of the mass pointcloud. We use a k-d tree structure to speed it up in the process of the closest point search as used in Greenspan et al. [31]. Also, we find that not all valid points are required to obtain a good alignment; instead, a sampling rate of 6 is used which greatly increases the matching efficiency. The average matching time is about 1.95 s per gallery face for all the six regions.

The selected regions can be referred to as Ω , and $\Omega \{A, B, C, D, E, F\}$. Given that multiple regions are matched separately, the final recognition results are fused using a weighted sum rule, and the confidence in each region is in inverse proportion to the deformation mapping in the corresponding region, which is given by

$$s_i = \sum_{\Omega} \omega_{\text{Probe}_M} * s_{\text{Probe}_M}, M \in \Omega, \quad (11)$$

$$\omega_{\text{Probe}_M} = \frac{1}{\sum_{\Omega} \frac{1}{d_{\text{Probe}_L}}}, L \in \Omega, \quad (12)$$

where ω_{Probe_M} is the weight of the region Probe M, and s_{Probe_M} is the similarity score of the region Probe M. In this way, the more deformation occurs in the region, the less confidence the region has. For each face in the probe set, the matching engine results in a similarity vector $\mathbf{s}$ of size N' , where N' is the size of the remaining galleries and $N' \ll N$. An element s_i of a vector $\mathbf{s}$ denotes the similarity score between the probe and gallery face number i . All the similarity vectors have a negative polarity, so the identity corresponding to the smallest similarity score s_i is supposed to be the recognition result.

Table 3
Facial regions information (mm).

Facial regions	Probe B	Probe C	Probe D	Probe E	Probe F	Gallery region
OffsetX	0	−35	35	−25	25	0
OffsetY	50	10	10	−35	−35	0
SphereRadius	25	25	25	25	25	85

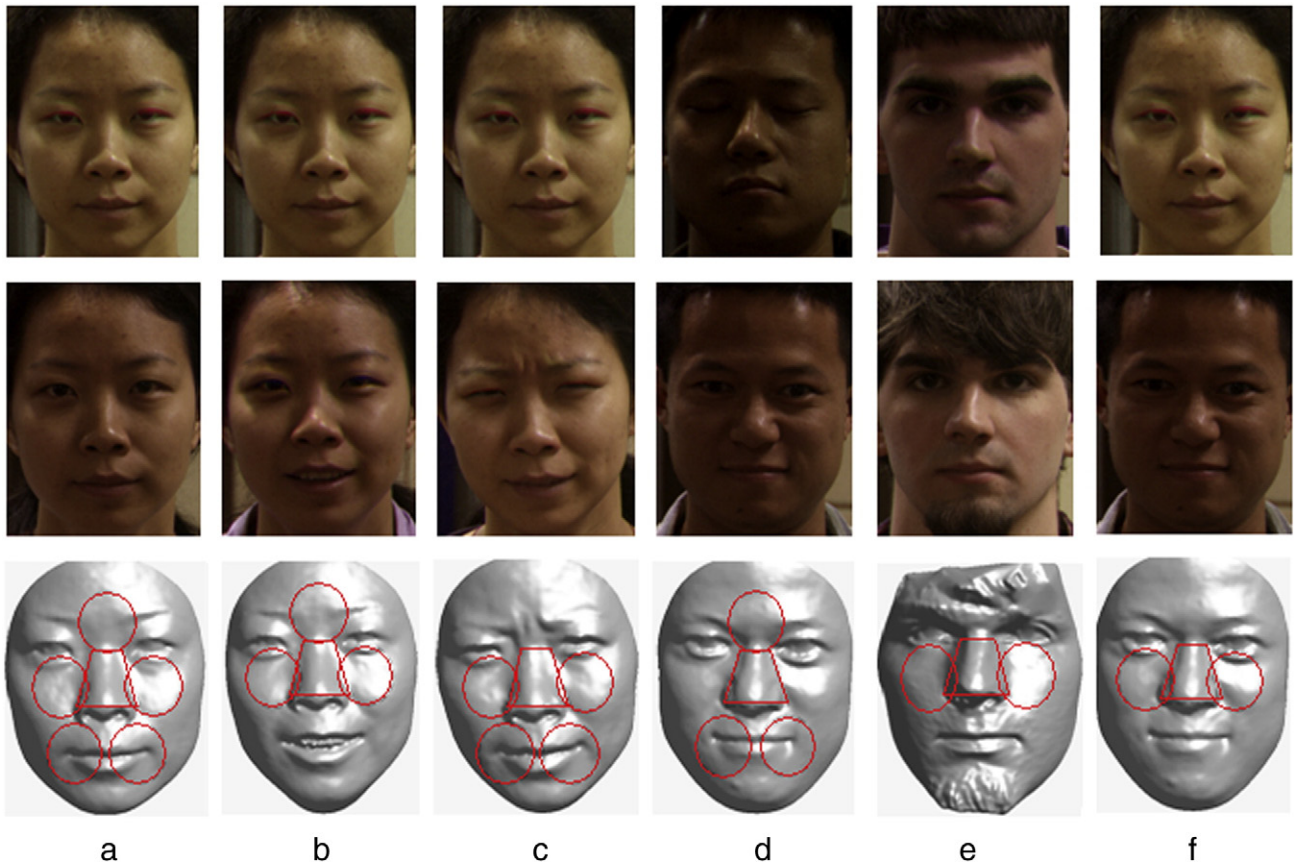


Fig. 10. Results of region selection. (a), (b), (c), (d), (e). Region selection when the matching faces belong to the same identity, and (f) region selection when faces belong to different identities. (Top: target face. Center: matching face. Bottom: results of region selection.)

5. Experiments and analysis

We employ two different scenarios for our experiments: verification and identification. In a verification scenario, the objective is to answer if a person is who he/she claims to be, and we measure the verification rate at a 0.1% false acceptance rate (FAR). Each face in the probe set is compared to a gallery set and the result is compared against a threshold. The results are summarized using receiver operating characteristic (ROC) curves. In an identification scenario, the objective is to answer who the person is to be, and we measure the identification rate by rank-one recognition. Each face in the probe set is compared against all gallery set, which is one-to-many matching. The system outputs a rank of the most similar faces based on the similarity score. The performance is measured using a cumulative match characteristic (CMC) curve.

5.1. Database description

The FRGC v2.0 dataset includes about 5000 3D faces under neutral and non-neutral expressions. The dataset is composed of two partitions: the training partition (943 scans) and the evaluation partition (4007 scans). The training scans are acquired in Spring 2003, and the evaluation partition scans belong to 466 subjects and are obtained in Fall 2003 and Spring 2004. Generally, there are more expressions and hair occlusions in the evaluation partition.

Since most of the scans in the training partition of FRGC are neutral, it is not sufficient for training the threshold δ_{deform} . So we augment it with the ZJU3DFED database [10]. In the ZJU3DFED database, there are 40 different subjects, nine scans for each, for a total of 360 scans. Each subject has two scans with a smiling expression, two scans

with a surprised expression, two scans with a sad expression and three scans with a neutral expression.

5.2. Parameter: δ_{deform}

The threshold δ_{deform} is the key for adaptive region selection scheme. It influences the recognition performance positively in one way and negatively in another way. On one hand, the larger the δ_{deform} is, the more regions with discriminative information can be used. This can result in a higher recognition performance. On the other hand, with a larger δ_{deform} there are more chances for introducing the information deformed by expression and hair occlusion, which will deteriorate the recognition performance.

The augmented training partition is used to train the threshold δ_{deform} . 50 subjects are randomly selected in the training set. One neutral scan of each subject is selected as the gallery face, the remaining scans of the 50 subjects constitute the probes and there are 347 scans for probes. Matching based on the adaptive region selection scheme is applied with δ_{deform} ranging from 0 to 2.5 mm. When δ_{deform} is 0, only the nose region is used for matching. With the increasing of the threshold δ_{deform} , more and more regions are selected. Except for the mouth regions with exaggerated expressions such as a largely opened mouth, almost all of the six regions are chosen when δ_{deform} is 2.5 mm. Their rank one recognition rates are shown in Fig. 11. With the increasing of δ_{deform} , the rank one recognition rate decreased a little, then it noticeably increased. Next, it is almost leveled at its best performance at the range of δ_{deform} from 0.9 to 1.1 mm. For a δ_{deform} larger than that, large performance degradations are noticed when δ_{deform} increases, and the instances of failure are mainly due to

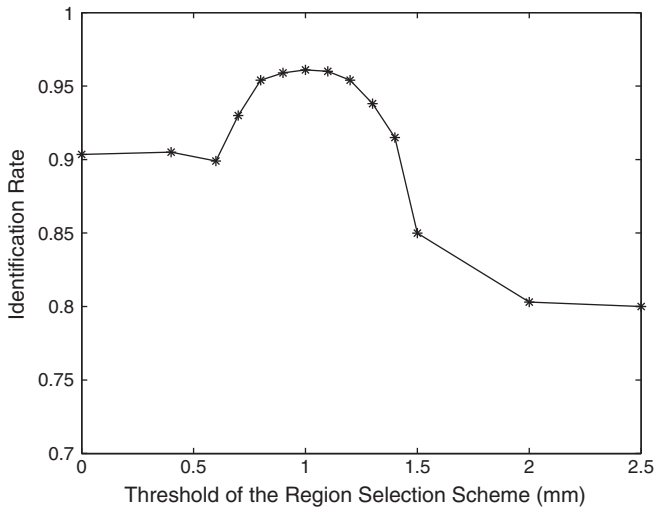


Fig. 11. Identification rate versus the value of the threshold δ_{deform} .

expression and hair occlusion. In the following experiments, we select δ_{deform} as 1.0 mm.

5.3. Time cost analysis

We implement our algorithm on an Intel Core Duo 2.34 GHz machine with 1.0 GB of memory. Table 4 shows the average running time of our algorithm in a verification scenario. From preprocessing to final decision, the verification process takes less than 9 s. This implies that a feasible execution time can be achieved for use at airport checkpoints, or bank ATMs, where the user may expect a slight authentication delay.

In an identification scenario, the rejection classifier takes 3.08 s for matching a probe with a gallery of 466 faces and only 144 faces remained to be processed in the next stage. The total execution time is $6.73 + 0.0066 \times 466 + (0.0528 + 1.23) \times 144 + 0.01 = 194.5388$ s with rejection, while it takes $6.73 + (0.0066 + 0.0528 + 1.23) \times 466 + 0.01 = 607.6004$ s without rejection. It is obvious that the recognition efficiency is improved greatly by the rejection classifier.

5.4. Verification experiment

In the verification experiment, we perform the FRGC ROC I, II and III. The ROC curves are presented in Fig. 12. Generally, the nose and forehead regions are supposed to be the steadiest regions, and are used for recognition. To test the performance of the adaptive region selection scheme, the performance of using the nose–forehead region is also presented in Fig. 12.

At a 0.1% FAR, our scheme can achieve the verification rates of 96.7%, 96.3% and 96.0% for ROC I, II and III, respectively. And the verification rates are 94.75%, 94.4% and 94.0% for using the nose and forehead regions. We can see that our scheme performs better. The results of our algorithm with rejection classifier are not reported

Table 4
Time cost analysis.

Step	Time (ms)
Data preprocessing	6730
Partial central profile matching	6.6
Other facial curves matching	52.8
Region-based matching	1230
Result fusion	10

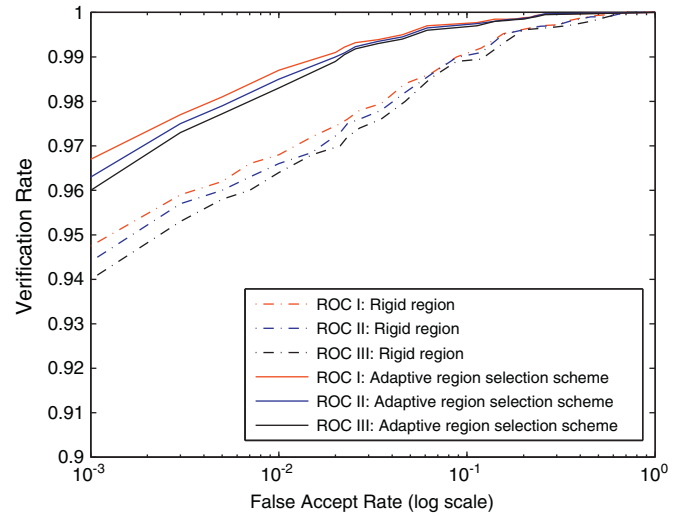


Fig. 12. ROC curve for the "Fall vs. Spring" experiment.

because the experiment is a one-to-one matching experiment, and there is no need to use a prior rejection.

We also perform the "All vs. All" experiment, that is, when the probe set (4007 faces) is matched with the gallery set (4007 faces). Our adaptive region selection scheme is able to achieve a verification rate of 95.3% at a FAR of 0.1%. Fig. 13 presents the corresponding ROC curve for this experiment.

5.5. Identification experiment

In the experiment, we take the earliest image of every participant and use it as the gallery image for the subject. All subsequent images for each subject are used as probes.

Fig. 14 shows the identification results of our algorithm. The results of using the nose–forehead region, regions adaptively selected and regions adaptively selected combined with rejection are presented. The rank-one recognition rates are 91.2%, 97.8% and 97.5%, respectively. It can be found that the identification rate with the rejection classifier is slightly reduced by 0.3% compared with the region-based method. The reason for this is that there is a probability that the classifier may also reject the correct identity. It can be avoided by improving

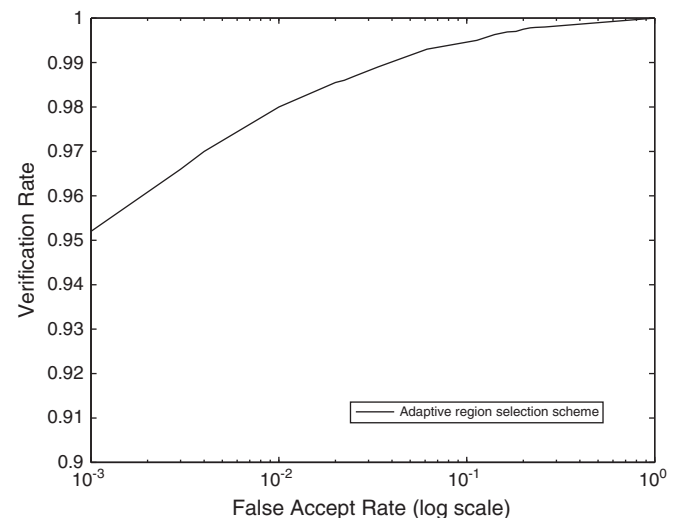


Fig. 13. ROC curve for the "All vs. All" experiment.

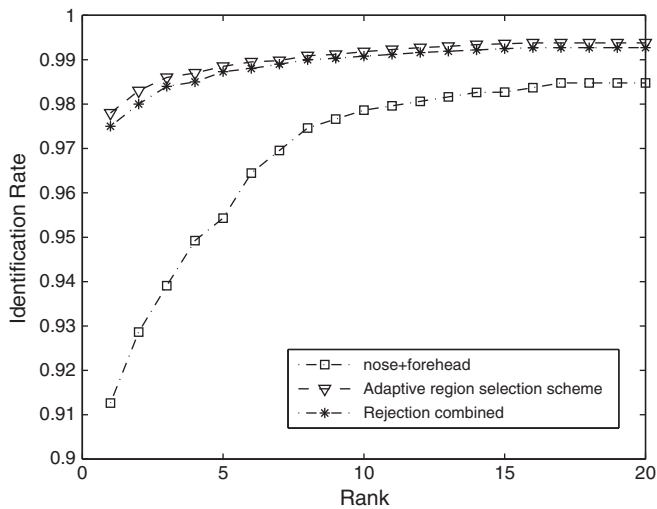


Fig. 14. CMC curves for the “First vs. Others” experiment.

the threshold δ_{rej} in rejection process. However, the selection of δ_{rej} should be balanced between the time consumption and recognition rate. And in many circumstances, a 69% improvement in runtime is more important than a loss of 0.3% accuracy.

To test the performance of our adaptive region selection scheme, here the performances of using the nose–forehead region, all 6 regions and regions adaptively selected in this paper are compared. The shape of the regions near the mouth is susceptible to expressions, so the 4 regions Probe A, Probe B, Probe C and Probe D are also used for recognition. For a fair comparison, all of the methods use sum rule for fusion. The rank-one recognition results are listed in Table 5.

We can see that our scheme performs best. The performance of the method that uses the nose and forehead regions is superior to the method that all 6 regions are used. This observation agrees with the claim in [7] that using less of the face can result in more accurate recognition and the reason for this is that expressions and hair occlusion may more likely occur in the larger frontal face region. However, we should mention that good recognition performance obtained by using less of the faces is mainly attributed to the adoption of the steady information in the face. In fact, the performance will be improved if more regions are used when all of them are not deformed. Our method performs better than the method which uses only the nose–forehead region which indicates that regions covering the cheeks and mouth indeed contain useful information for face recognition, and the forehead region is no longer steady in the presence of certain expressions and hair occlusion.

5.6. Fusion scheme

When the regions used for matching are automatically selected, results for recognition can be improved by employing the fusion method based on weighted sum rule. Table 6 shows a comparison between performance when using the sum rule and the weighted sum rule.

Table 5
Identification results.

Features	Rank-one recognition rate
Nose–forehead region	91.2%
4 regions (forehead, nose and cheek regions)	87.7%
All 6 regions	78.9%
Regions adaptively selected	97.4%

Table 6
Verification rate at 0.1% FAR.

Datasets	Sum rule	Weighted sum rule
Fall vs. Spring	95.4%	96.0%
All vs. All	94.2%	95.3%

Table 7
Identification results using the “First vs. Others” dataset.

Method	Rank-one
Cook et al. [16]	92.0%
Kakadiaris et al. [4]	97.0%
Faltemier et al. [11]	97.2%
Queirolo et al. [9]	98.4%
This paper	97.8%

Based on these experimental results one can see that our fusion scheme based on the weighted sum rule produces better results when compared with the fusion scheme based on the sum rule.

5.7. Comparison with other methods

Some authors have published their results using the FRGC v2.0 dataset. We compare our results with theirs on the same database. Table 7 presents the rank-one results using the First vs. Others dataset, and results from FRGC ROC III are available in Table 8. Table 9 shows a verification experiment using the All vs. All dataset.

Note that our algorithm outperforms each method presented except that of Kakadiaris et al. [4] and Queirolo et al. [9]. Kakadiaris et al. present an algorithm for 3D face recognition that uses an AFM to create a unique representation of a human face. This algorithm first fits an AFM to a 3D face, then the fitted AFM is parameterized to produce three channel deformation images, next wavelet transforms are used to obtain the features of the 3D face, and the features are utilized for recognition in the end. Their algorithm can be thought of as a non-rigid algorithm, and our algorithm belongs to a rigid one. The non-rigid algorithm is supposed to be more promising in handling the expression challenge [3], but it is usually complex and susceptible to occlusions which are typical in realistic applications. Compared with Kakadiaris’ algorithm, we use a simpler rigid approach that does not require an AFM and can handle occlusions. Both Queirolo et al.’s method and ours belong to a rigid algorithm. SA algorithm is used for matching in their paper, and it is a stochastic algorithm for local search which can avoid local minima. ICP algorithm is used in our paper and ICP algorithm

Table 8
Verification results for FRGC ROC III, at 0.1% FAR.

Method	Verification rate
Al-Osaimi et al. [6]	94.1%
Faltemier et al. [11]	94.8%
Queirolo et al. [9]	96.6%
Kakadiaris et al. [4]	97.0%
This paper	96.0%

Table 9
Verification results using the “All vs. All” dataset.

Method	Verification rate
Mian et al. [8]	86.6%
Cook et al. [16]	92.3%
Faltemier et al. [11]	93.2%
Queirolo et al. [9]	96.5%
This paper	95.3%

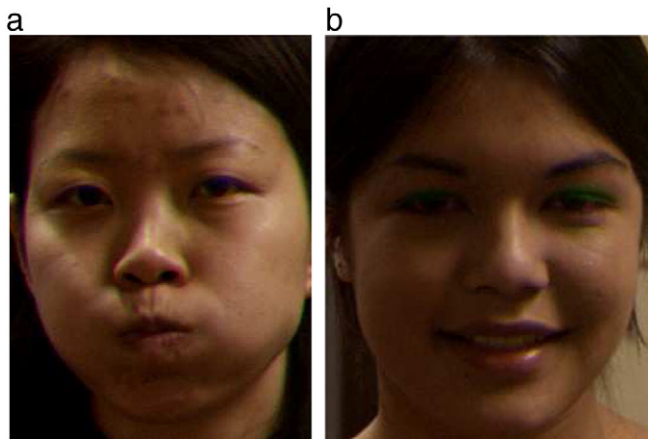


Fig. 15. Failure cases. (a) Failure case caused by incorrect normalization, and (b) failure case caused by expression.

tends to converge to local minimum solutions, so Queirolo et al. obtain the better performance. However, it is well known that SA is time consuming and 1.5 s is reported for the average execution time of the nose circle using SA. We perform the same experiment using ICP, and only 0.6 s is needed. Also, six feature points containing the nose tip need to be detected in their paper, while just the nose tip is needed in our paper which makes the method simpler. Besides, rejection classifier can reduce 69% dissimilar faces in the gallery before fine registration in our paper.

6. Conclusions

In this paper, we have proposed a robust 3D face recognition method based on rejection and adaptive region selection scheme. Experiments have been done on the FRGC v2.0 dataset. Encouraging results and comparisons to other 3D face recognition method have demonstrated the effectiveness of our method.

Although our method works well on common faces with approximately symmetrical shapes, it can fail when the face is normalized incorrectly, which causes incorrect PCS. Some failure cases are provided in Fig. 15. The accurate location of the nose tip is vital for the pose normalization. However, the nose detection will fail due to hair covering a part of the face and in a few cases due to exaggerated expressions (e.g. widely open mouth and inflated cheeks, see Fig. 15(a)). Fortunately, this is a rare case and most images in the FRGC v2.0 dataset can be handled by our algorithm. As mentioned before, the deformation mapping in this paper contains intra-personal deformation and inter-personal deformation when the matching faces belong to different identities. Hence regions with large expressions or hair occlusion will be selected if the deformation can be compensated by inter-personal deformation which will deteriorate the recognition performance. The performance will be improved if inter-personal deformation can be accurately separated. We will focus on separating the inter-personal deformation from the intra-personal deformation in our future work.

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